Airbnb Market Analysis Project

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Project Type: Data Analytics & Business Intelligence Project

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1. Project Overview

The **Airbnb Market Analysis Dashboard** project focuses on analyzing Airbnb listing data to identify pricing trends, booking behavior, and revenue potential across different neighborhoods.

Using **Python for data cleaning and analysis** and **Power BI for visualization**, the project transforms raw Airbnb datasets into meaningful insights that can support business decisions for hosts, investors, and market analysts.

The final outcome is an **interactive dashboard** that provides a clear view of the Airbnb marketplace and helps stakeholders identify high-performing locations and pricing strategies.

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2. Problem Statement

Airbnb hosts and property investors often face difficulty identifying the most profitable locations, optimal pricing strategies, and high-demand property types.

Without proper data analysis, decisions related to property investment, pricing, and market positioning may rely on assumptions rather than data-driven insights.

This project aims to solve this problem by analyzing Airbnb datasets to uncover:

- Pricing patterns across neighborhoods
- Booking trends across listing types
- Revenue potential across locations
- Host performance and its impact on bookings

The insights generated from this analysis can help stakeholders make **data-driven decisions to improve profitability and market performance.**

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3. Project Objective

The objective of this project is to analyze Airbnb listing data to understand:

- Pricing trends across neighborhoods
- Distribution of listing types
- Booking behavior of customers

• Revenue potential across locations
• Impact of host status on bookings

The final outcome is an **interactive Power BI dashboard** that enables stakeholders to explore Airbnb market performance and identify profitable areas.

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4. Tools & Technologies Used

Category	Tools
Data Cleaning	Python
Libraries	Pandas, NumPy
EDA	Matplotlib
Dashboard	Power BI
Environment	Jupyter Notebook
Data Formats	CSV, JSON

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5. Data Pipeline

The project follows a structured **data analytics workflow** to transform raw data into meaningful business insights.

Data Pipeline Process

Raw Airbnb Data (Listings, Calendar, Reviews, Neighborhood JSON)
↓
Data Cleaning using Python Handling missing values, formatting columns, and removing unnecessary fields
↓
Feature Engineering Creating new metrics such as booking rate and availability rate
↓
Exploratory Data Analysis (EDA) Identifying trends in pricing, listings, and booking behavior

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Data Integration Merging multiple datasets using the **listing_id** key
↓
Dashboard Development Building interactive visualizations using **Power BI**
↓
Business Insights Generating insights and recommendations for Airbnb market performance

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6. Dataset Overview

The project used multiple Airbnb datasets which were cleaned and merged to create a unified analytical dataset.

Final Dataset Structure

Metric	Value
Total Records	18,017
Unique Listings	6,489
Total Columns	83

Key Variables Used in Analysis

Column	Description
listing_id	Unique listing identifier
room_type	Type of accommodation
price	Price per night
neighbourhood_cleansed	Listing location
availability_365	Annual availability

review_scores_rating	Customer rating
estimated_revenue_l365d	Estimated yearly revenue
booking_rate	Calculated booking ratio

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7. Data Cleaning Process

Data cleaning was performed using **Python (Pandas)** in **Jupyter Notebook** to prepare the Airbnb dataset for analysis and dashboard development.

The cleaning process was applied to the following datasets:

- Listings Dataset
- Calendar Dataset
- Reviews Dataset
- Neighbourhood Dataset

7.1 Listings Dataset Cleaning

The Listings dataset was the **primary dataset** used in the analysis. It contained detailed information about Airbnb properties, hosts, pricing, location, and review metrics. **Initial Dataset Structure**

Metric	**Value**
Total Records	18,017
Total Columns	83
Data Types	object, float64, int64, datetime

Important Columns

- listing_id
- host_id
- room_type
- neighbourhood_cleansed
- price
- availability_365
- review_scores_rating

• estimated_revenue_l365d

Missing Value Analysis

Column	**Observation**
host_name	Few missing values
host_since	Few missing values
Column	**Observation**
host_response_time	Some missing values
host_response_rate	Some missing values
host_acceptance_rate	Some missing values
host_neighbourhood	Large number of missing values
neighbourhood	Missing values present
license	Some missing values

Cleaning Actions Taken

1. Irrelevant columns were ignored

Columns not required for analysis were excluded.

Examples include:

• picture_url
• listing_url
• host_url

These columns do not contribute to analytical insights.

2. Datetime columns were standardized

The following columns were converted into datetime format:

• last_scraped
• host_since

• first_review
• last_review

This allows time-based analysis.

3. Price column formatting

The price column was converted into **numeric format** to ensure accurate calculations in analysis and Power BI visualizations.

4. Handling missing categorical values

Columns like:

- host_response_time
- host_is_superhost • neighbourhood were retained but treated as

categorical variables during analysis.

5. Verification of unique listings

To confirm the number of unique properties:

df_final['listing_id'].nunique()

Result:

6,489 unique listings

This indicates that some listings have **multiple observations across datasets after merging.**

7.2 Calendar Dataset Cleaning

The calendar dataset contained information about **daily listing availability and price**.

Key Cleaning Steps

• Removed unnecessary columns not required for analysis
• Converted price values into numeric format

Created Availability Metrics

Several metrics were calculated:

• Available Days
• Total Days
• Availability Rate

Booking performance metric:

Booking Rate =

1 − Availability Rate

Example Output

listing_id	avg_price	available_days	total_days	availability_rate	booking_rate
11156	65	364	365	0.997	0.002
15253	99	216	365	0.591	0.408
listing_id	**avg_price**	**available_days**	**total_days**	**availability_rate**	**booking_rate**
44545	130	5	365	0.013	0.986

These metrics help estimate **occupancy and demand for listings.**

7.3 Reviews Dataset Cleaning

The reviews dataset contained **guest feedback information**. **Missing Value Analysis**

Column	Missing Values
English	2
listing_id	0
id	0
date	0
reviewer_id	0
reviewer_name	0

Cleaning Action

The missing review values were **removed**, since they represented a very small portion of the dataset.

This ensured **clean review text data for analysis.**

7.4 Neighbourhood Dataset

The neighbourhood dataset was provided in **JSON format**.

After inspection, the dataset did not require significant cleaning because:

• Data structure was consistent
• No missing values affecting analysis
• Geographic information was standardized

Therefore, it was used **directly for merging with listing data.**

7.5 Data Merging

After cleaning each dataset individually, they were merged using the key column:

listing_id

The final dataset combined information from:

• Listings dataset
• Calendar dataset
• Reviews dataset • Neighbourhood dataset

Result:

Final dataset contained **83 columns and 18,017 rows.**

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8. Exploratory Data Analysis (EDA)

EDA was conducted to understand **distribution patterns and relationships within the data.**

8.1 Room Type Distribution

Room Type	Listings
Entire home/apt	4902
Private room	1497
Hotel room	86
Shared room	4

Insight

Entire homes dominate the Airbnb market, representing the majority of listings. This indicates that most hosts offer **full property rentals instead of shared accommodations. 8.2 Average Price by Room Type**

Room Type	Average Price
Entire home/apt	371
Hotel room	262
Private room	123
Shared room	74

Insight

Entire homes command the highest price while shared rooms represent the **lowest-cost segment**.

8.3 Average Price by Neighbourhood

Top neighbourhoods by price:

Neighbourhood	Avg Price
Pittwater	655
Woollahra	611
Mosman	512
The Hills Shire	401
Manly	388

Insight

These areas represent **premium Airbnb markets** with high property values.

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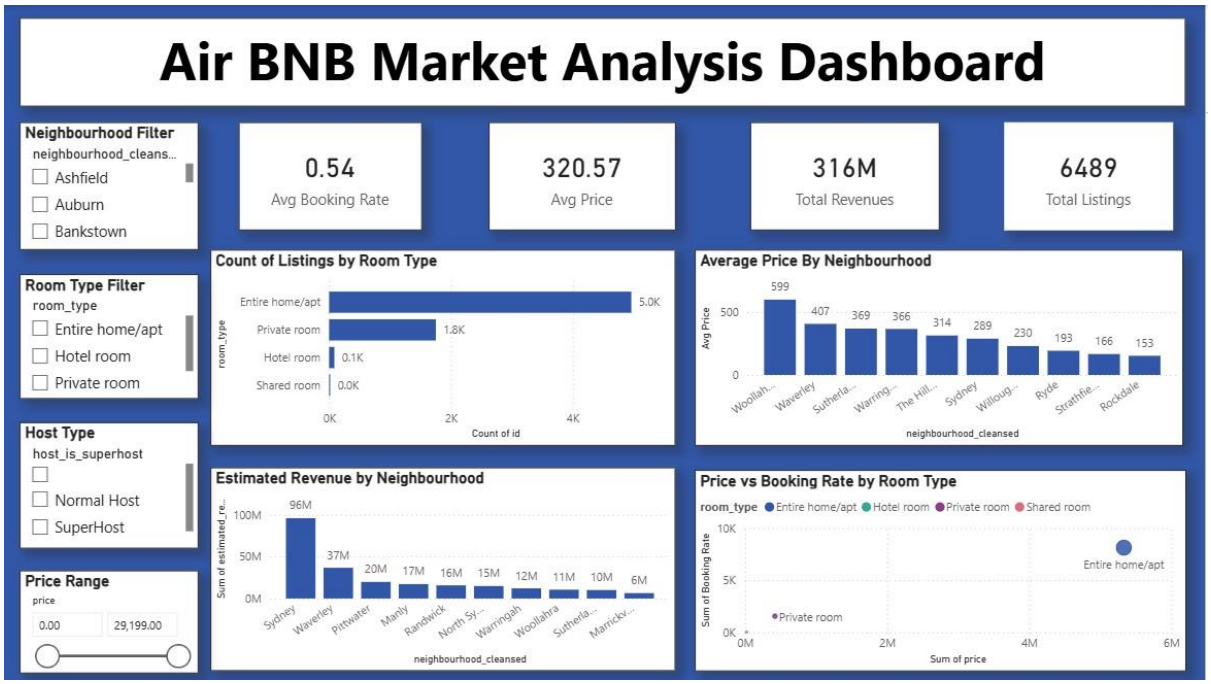
9. Power BI Dashboard



The final analysis was presented using an **interactive Power BI dashboard**.

The dashboard allows users to explore:

- Listings distribution
- Pricing patterns
- Revenue performance
- Booking behavior

9.1 Dashboard Components

KPI Cards

KPI	Value
Total Listings	6,489
Avg Price	320.57
Avg Booking Rate	0.54
Total Revenue	316M

9.2 Visualizations Used

1. Listings by Room Type

Shows the distribution of listing types.

2. Average Price by Neighbourhood

Identifies premium pricing areas.

3. Estimated Revenue by Neighbourhood

Highlights top revenue generating locations.

4. Price vs Booking Rate

Analyzes relationship between pricing and occupancy.

9.3 Dashboard Filters

Interactive filters were added for dynamic exploration.

Filter	Purpose
Neighbourhood	Location analysis
Room Type	Listing segmentation
Host Type	Superhost vs Normal host
Price Range	Budget vs premium listings

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10. Key Business Insights

Market Structure

Entire homes represent the majority of Airbnb listings.

Pricing Trends

Premium neighbourhoods command significantly higher prices.

Revenue Concentration

A small number of neighbourhoods generate large portions of revenue.

Booking Behavior

Higher priced listings may experience **lower booking rates**.

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11. Strategic Recommendations

Based on the analysis, the following strategies are recommended.

Dynamic Pricing

Hosts should adjust prices based on demand and booking trends.

Location-Based Investment

Property investments should focus on **high-performing neighbourhoods**.

Market Segmentation

Hosts should offer different pricing tiers to target **luxury and budget travelers**.

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12. Conclusion

This project demonstrates how **data analytics and business intelligence tools** can transform raw Airbnb data into actionable insights.

By combining **Python for data analysis** and **Power BI for visualization**, the project delivers a comprehensive view of Airbnb market dynamics.

The dashboard enables stakeholders to make **data-driven decisions regarding pricing strategies, location investment, and listing performance.**